INCH-POUND

MIL-PRF-49063E
4 March 1999
SUPERSEDING
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21 May 1992

PERFORMANCE SPECIFICATION
NIGHT VISION SIGHT, INDIVIDUAL SERVED WEAPON AN/PVS-4

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Night Vision Sight, Individual Served Weapon, AN/PVS-4.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-P, Fort Monmouth, NJ 07703-5023 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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2.2 Government Documents

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of this documents are those listed in the Department of Defense Index of Specifications and Standards DODISS) and supplement thereto, cited in the solicitation.

Specifications
MILITARY

MIL-PRF-49040	-	Image Intensifier Assembly, 25 Millimeter, Microchannel Inverter, MX-9644/UV
MIL-PRF-49471/4	-	Battery, Non-rechargeable, High Performance, BA-X567/U

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Document Order Desk, 700 Robbins Ave, Bldg #4, Section D, Philadelphia, PA 19111-5094).

2.2.2 Reference documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2.i).

DRAWINGS

USA COMMUNICATIONS-ELECTRONICS COMMAND

SM-D-850300	-	Night Vision Sight, Individual Served Weapon AN/PVS-4
SM-D-850482	-	Bag, Carrying

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2.3 Non-Government Publications The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of documents cited in the solicitation (see 6.2.i).

ANSI C18.1M Batteries and Dry Cells Specifications
Specification Sheet 20-0571-1988

(Application for copies should be addressed to the American National Standards Institute , 11 West 42nd Street, New York, NY 10036)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The Night Vision Sight, Individual Served Weapon AN/PVS-4, herein, referred to as the sight, is intended for use at night for aimed firing of individual served weapons. The sight assembly consists of an objective lens assembly, battery housing assembly, an eyepiece assembly, an eyeguard assembly, AA Battery Adapter for BA 3058 batteries, an image intensifier assembly, a daylight cover, a carrying and storage case, a carrying bag and ancillary equipment consisting of a brush, lens tissue, M60 reticle cell assembly, M60 mount and M16 mounting knob assembly (see 2.1 and 2.2).

3.2 First article testing (FAT). When specified in the contract (see 6.2.d and 6.2.g), the contractor shall furnish FAT sights for inspection and approval (see 4.2).

3.3 Construction. The sight shall be constructed to assure compliance with the contract configuration baseline and in accordance with SM-D-850300 as applicable.

3.4 Material. The contractor shall select parts and materials that are fully capable of meeting all of the operational and environmental requirements specified herein. The materials specified in the applicable drawings are recommended , but not mandatory. Selection criteria of the class, grade or type part shall ensure the material will be able to perform its intended function when it is assembled. Verification of the supplier meeting the overall performance requirements shall be the governing acceptance standard. Recovered materials shall be used to the maximum extent possible.

3.5 Major components. Requirements for conformance inspection of major components shall be as specified in the contract or purchase order (see 6.2.f).

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3.6 Performance Characteristics.

3.6.1 Sight definition. For conformance to the following performance parameters, the sight is defined as sight less case, bag and ancillary equipment.

3.6.2 Eyeguard operation. With the eyeguard vertical and at -48°C, the eyeguard shall open (depress) when a 2000, ± 10 gram weight is placed axially on the eye flange and shall close (expand) when the weight is reduced to 100, ± 10 grams. At +23°C, the eyeguard shall open (depress) when a 1600, ± 10 gram weight is placed axially on the eye flange and shall close (expand) when the weight is reduced to 100, ± 10 grams. At +23°C and with the eyeguard open (depressed) the minimum open position of the 2 leaves shall be 1.875 inches.

3.6.3 Resolution. The minimum sight resolution shall be within one element number of the resolution of the image intensifier used in the sight at the time of testing while using a USAF 1951 Resolving Power Test Target with the target brightness adjusted to achieve best contrast.

3.6.4 Field of View. The sight shall have a minimum field of view of 253 milliradians.

3.6.5 Weight. The weight of sight with BA-5567/U battery but without daylight cover shall not exceed 4.0 pounds.

3.6.6 Controls and adjustments. All controls and adjustments shall operate smoothly without binding throughout their complete range of movement at all operating temperatures specified herein.

3.6.6.1 Eyepiece diopter focus. With the diopter adjustment control set at -6 (+ 1/4, -0.0) to +2 diopter, the image of the projected reticle shall be in sharp focus.

3.6.6.2 Range focus. The sight shall provide a minimum focus adjustment range from a near focus of 25 meters to infinity. The dynamic torque to turn the range focus control shall be 8 ± 7 inch-pounds.

3.6.6.3 Tube brightness control. The tube brightness control shall vary the gain of the image intensifier tube from OFF to maximum gain.

3.6.6.4 Reticle brightness control. Brightness of the reticle pattern shall be continuously variable over the range from OFF to full brightness. The pattern shall be visible and the numbers legible when viewing a solid white target having a brightness of 3.3×10^{-3} footlamberts minimum.

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3.6.6.5 Reticle movement. Two external control knobs shall control the position of the reticle pattern viewed by the observer using the sight. One control shall move the reticle pattern in azimuth; another control shall move the reticle pattern in elevation. An adjustment made by turning one control shall not cause the reticle to deviate from a straight line of travel by more than 0.5 milliradian. Adjustment will be apparent to the operator by tactile indexing with audible clicks. Each click of the control knob shall displace the reticle pattern 0.25, ± 0.04 milliradian. Total movement of the reticle pattern shall be a minimum of 2.5° in any direction from the optical axis. The torque required to adjust the reticle position shall be between 0.5 and 4.0 inch-pounds. The reticle pattern shall remain in focus throughout the full range of movement of the range focus control. The reticle pattern shall not exhibit rotation greater than 1° from its proper orientation.

3.6.6.6 Battery adapter. The sight shall properly operate with the battery adapter attached and Type BA-3058/U batteries conforming to ANSI C18.1M, Specification Sheet 20-0571-1988 inserted. The daylight cover may be installed during this test. Observe the sight for proper illumination.

3.7 Environmental characteristics.

3.7.1 Temperature extremes. The sight shall not be damaged (see 3.10.1) by storage in any temperature from +23°C to +68°C or from +23°C to -51°C or by operation at any temperature from +23°C to +52°C or from +23°C to -51°C. The sight shall not be damaged (see 3.10.1) after being subjected to repetitive temperature changes between +23°C and +68°C in a 5 minute period and between +23°C and -51°C in a 5 minute period. Soak time at temperature extremes shall be sufficient for the entire sight to attain stabilization. Time for temperature stabilization is defined as the amount of time required for the most internal part of the unit under test to achieve the specified temperature. Validation of operation shall be observance of a green glow at the eyepiece at operating temperature and, at low operating temperature, the ability to operate all controls with arctic gloves or equivalent.

3.7.2 Vibration. The sight shall not be damaged (see 3.10.1) after being subjected to simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion) with the frequency being varied between 5 and 55 hertz (Hz). Operation of the sight during vibration shall not be required. Vibration shall be applied in 3 mutually perpendicular planes, one of which is perpendicular to the optical axis. The duration of vibration shall be 5 minutes in each plane. The test after vibration shall be resolution.

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3.7.3 Shock.

3.7.3.1 High intensity shock. The sight shall not be damaged (see 3.10.1) by a sequence of 3 shocks applied in each direction along each of 3 mutually perpendicular axes (horizontal, vertical, and optical axes when the reticle pattern is erect) with the sight in the non-operating mode. The shocks shall be half sine pulses and shall have a time duration of 4 milliseconds, ± 5 percent. Shock pulses along the optical axis shall have a peak amplitude of 100 g's ± 15 g's (see 3.10.3).

3.7.3.2 Weapons Shock. The sight, mounted and zeroed (see 3.10.2) on the M16 Rifle, shall not shift in zero more than ±0.50 milliradians or be damaged (see 3.10.1) when subjected shocks encountered during rifle firing.

3.7.4 Altitude. The sight shall not be damaged (see 3.10.1) when operated (see 4.4) at a pressure equivalent to 10,000 feet altitude above sea level. The sight shall be operated for a period of not less than 3 hours.

3.7.5 Humidity. The sight shall not be damaged (see 3.10.1) after being operated (see 4.4) in an atmosphere with a relative humidity of 95 percent, ± 5 percent over a temperature range of +20°C to +52°C and shall operate after storage up to +68°C minimum with 95 percent, ± 5 percent relative humidity.

3.7.6 Immersion. The sight shall not be damaged (see 3.10.1) after being immersed in fresh water to a depth of not less than 3 feet for a period of not less than 30 minutes. The sight shall be pre-conditioned by exposure to temperature of +45°C ±3°C for a period of 2 hours prior to being immersed in the fresh water at a temperature of +18°C ±5°C.

3.7.7 Interchangeability. The objective lens assembly and the eyepiece lens assembly shall be physically interchangeable between sight assemblies without degradation of the resolution (see 3.6.3), eyepiece diopter focus (see 3.6.6.1), and range focus (see 3.6.6.2) requirements. These interchanges shall be made without any modification or adjustment beyond that necessary during replacement of any of the parts or assemblies involved.

3.8 Marking. Marking shall be in accordance with applicable drawings unless otherwise specified in the contract.

3.9 Workmanship. Standards of workmanship shall be such that the sight will meet all requirements of this specification and any referenced specifications or drawings.

3.10 Technical interpretations. The following technical interpretations, when referenced in sections 3 or 4, are mandatory for this specification.

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3.10.1 Damage. Breakage, loosening, shifting, evidence of corrosion or failure of any finish, hardware, connection or component; leakage or condensation of moisture within the sight; degradation in input or output characteristics; and any failure of the control knobs to perform their function.

3.10.2 Zero. A known and specified relationship between the sight aiming point and the strike of the bullet.

3.10.3 "g". "g" is defined as an acceleration or deceleration of 32.17 feet per second per second.

3.10.4 Room temperature. Room temperature is defined as +23°C, ± 10°C.

3.10.5 Environmental temperature. Environmental chamber temperatures shall be controlled to within ± 5°C of the specified temperature unless otherwise specified.

3.10.6 Illumination levels shall be ±10% of the light level specified.

3.10.7 Tolerance on 2854K color temperature shall be ±200K.

4. VERIFICATION

4.1 Classification of inspection. Inspection shall be classified as follows:

a. First article test (4.2)
b. Conformance inspection (4.3).

4.2 First article test (FAT). When specified in the contract or purchase order (see 6.2.d), first article testing shall be performed by the contractor. The quantity of first article test samples shall be as specified in the contract or purchase order (see 6.2.g) and shall be selected from the first production lot.

4.2.1 FAT testing. Each sight in the FAT lot shall be subjected to the inspections of Table I. Samples selected from the FAT lot shall be subjected to the inspections of Table II. Sample size shall be specified in the contract or purchase order (see 6.2.g). Failure of any inspection shall be cause for rejection of that sight and may be cause for failure of the FAT. Inspections may be conducted in any order except for Immersion, which shall follow temperature extremes and temperature shock.

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TABLE I. First article testing unit test.

Inspection	Requirement Paragraph
Resolution	3.6.3
Field of view	3.6.4
Eyepiece focus	3.6.6.1
Range focus	3.6.6.2
Tube brightness	3.6.6.3
Reticle brightness	3.6.6.4
Reticle movement	3.6.6.5
Battery adapter	3.6.6.6

TABLE II. First article testing sample test.

Inspection	Requirement Paragraph
Eyeguard operation	3.6.2
Temperature extremes	3.7.1
Vibration	3.7.2
High intensity shock	3.7.3.1
Weapons Shock	3.7.3.2
Altitude	3.7.4
Humidity	3.7.5
Immersion	3.7.6
Interchangeability	3.7.7
Weight	3.6.5

4.2.2 Disposition of FAT samples. After successful completion of the FAT, the FAT sample units shall be refurbished and delivered in accordance with the contract or purchase order (see 6.2.g).

4.3 Conformance inspection. Requirements for conformance inspection shall be as specified in the contract or purchase order (see 6.2.f).

4.4 Operational test. The test of the sight required during environmental testing shall consist of power up of the sight and observance of the green glow at the eyepiece.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2.k). When actual packaging of materials is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The AN/PVS-4 is intended for use at night for aimed firing of individual served weapons.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Which of the following major components are to be purchased and the quantities involved:
   M60 Mounting bracket per SM-D-850340
   M67 Mounting bracket per SM-D-850350
   M72 Mounting bracket per SM-D-850360
   M79 Mounting bracket per SM-D-850370
   M16 Mounting knob per SM-D-850500
   Bag, Cloth, Laminated Vinyl-Nylon SM-D-850482-1
   Bag, Cloth, Nylon (for Arctic Use) SM-D-850482-2
   Reticle M60, M14 per drawing SM-D-850490-2
   Reticle M67 per drawing SM-D-850490-4
   Reticle M72Al per drawing SM-D-850490-5
   Reticle M16, M16/XM203, M79 per drawing SM-D-850490-7
c. If the image intensifier tube per MIL-PRF-49040 is to be purchased:
   (1) To obtain the sight with Image Intensifier Tube, specify the use for drawing SM-D-850100-1
   (2) To obtain the sight without Image Intensifier Tube, specify the use of drawing SM-D-850100-2
d. Statement specifying that FAT testing is, or is not, required and sampling requirements for FAT.
e. Necessary actions by the contractor in the event of a lot test failure.
f. Statement identifying that the contractor must submit his proposal for conformance inspection to include the specific requirements to be verified and the sampling plan to be used to provide the desired confidence level.
g. Quantity and schedule for FAT testing for all line items and disposition of FAT samples.

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h. Levels of preservation and packaging (see section 5).
i. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).
j. Statement identifying if batteries are to be supplied.
k. Packaging and preservation (see section 5).

6.3 Definitions. See 3.10.

6.5 Subject term keyword listing.
Image intensification
Machine Gun Sight

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensive changes.

Custodian:
Army - CR
Air Force - 99

Preparing activity:
Army- CR

Project No. 5855-0103

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-49063E	**2. DOCUMENT DATE (YYMMDD)** 990304

3. DOCUMENT TITLE
Night Vision Sight, Individual Served Weapon AN/PVS4

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME US Army Communications-Electronics Command	b. TELEPHONE (Include Area Code) (1) Commercial (732) 532-9129 (2) AUTOVON 992-9129
c. ADDRESS (Include Zip Code) ATTN: AMSEL-LC-LEO-E-EP Fort Monmouth, NJ 07703-5023	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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